Mid Semester Examination 2025

Name of the Program: B.Tech Petroleum Engineering — *Semester: III*
Name of the Course: Thermodynamics — *Course Code:* TPE-301

Time: 1-1/2Hour — Maximum Marks: 50

Note:

i. Answer **all the questions** by choosing ***any one of the sub questions***.
ii. Each question carries 10 marks

Q1	(10 marks)	
(a)	What do you mean by a system, a closed system, an open system and an isolated system, boundary, Imaginary boundary and surroundings? Differentiate homogeneous and heterogeneous system with examples.	CO 1, CO 2
	OR	
(b)	**State** weather the following properties are intensive or extensive: i. Specific Volume ii. Density iii. Specific gravity iv. Heat capacity v. Specific heat vi. Internal energy vii. Pressure viii. Temperature ix. Specific heat capacity x. Mass	
Q2	(10 marks)	
(a)	A volume of 0.398 m^3 of gas at 1.03bar and 288K is compressed reversible and adiabatically to 10 bar. It is then cooled at constant pressure to its original temperature after which it expands isothermally. Calculate the following: i. Heat transfer at constant pressure ii. The heat transfer during expansion. iii. The total work done during the cycle.	CO 2
	OR	
(b)	Define the following: i. Closed, open, and isolated system ii. Intensive and Extensive Properties with examples iii. Degree of freedom iv. Homogeneous and heterogeneous system v. First law of thermodynamics	
Q3	(10 marks)	
(a)	Illustrate Isobaric, Isothermal and Adiabatic process and hence derive an expression for the work done and internal energy under isothermal condition of the system. Specify the symbol used	CO 1
	OR	
(b)	Air at 1.02 bar, 22^{0}C, initially occupying a cylinder volume of 0.015 m3, is compressed reversibly and adiabatically by a piston to a pressure of 6.8 bar. Calculate	

	i. The final temperature; ii. The final volume; iii. The work done	
Q4	(10 marks)	
(a)	Establish the relationship between PV, TV and PT for an adiabatic process.	**CO 1, CO 2**
	OR	
(b)	Explain compressibility factor (Z)? Discuss compressibility charts with the help of Z-P diagram for the real gases.	
Q5	(10 marks)	
(a)	A system consisting of a gas confined in a cylinder undergoing the following series of processes before it is brought back to the initial conditions: *Step 1: Isobaric process when it receives 50J of work and gives up 25J of heat* *Step 2: Isochoric process when it receives 75J of heat.* *Step 3: An adiabatic process* Determine the change in internal energy during each step and the work done during the adiabatic process.	**CO 1, CO 2**
	OR	
(b)	Heat is transferred to 10 kg of air which is initially at 100kPa and 300 K until its temperature reaches 600 K. Determine the change in internal energy, the change in enthalpy, and the heat supplied, and the work done in the following processes: i. Constant Volume process ii. Constant Pressure process	